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AIML

B-19

BDA TA1

[Github](#)

[Colab URL](#)

Problem Statement

With millions of tweets generated daily, manually analyzing public sentiment is impossible. This project applies Big Data Analytics to classify 1.6 million tweets as positive or negative using Apache Spark MLlib, demonstrating scalability of distributed ML on commodity hardware.

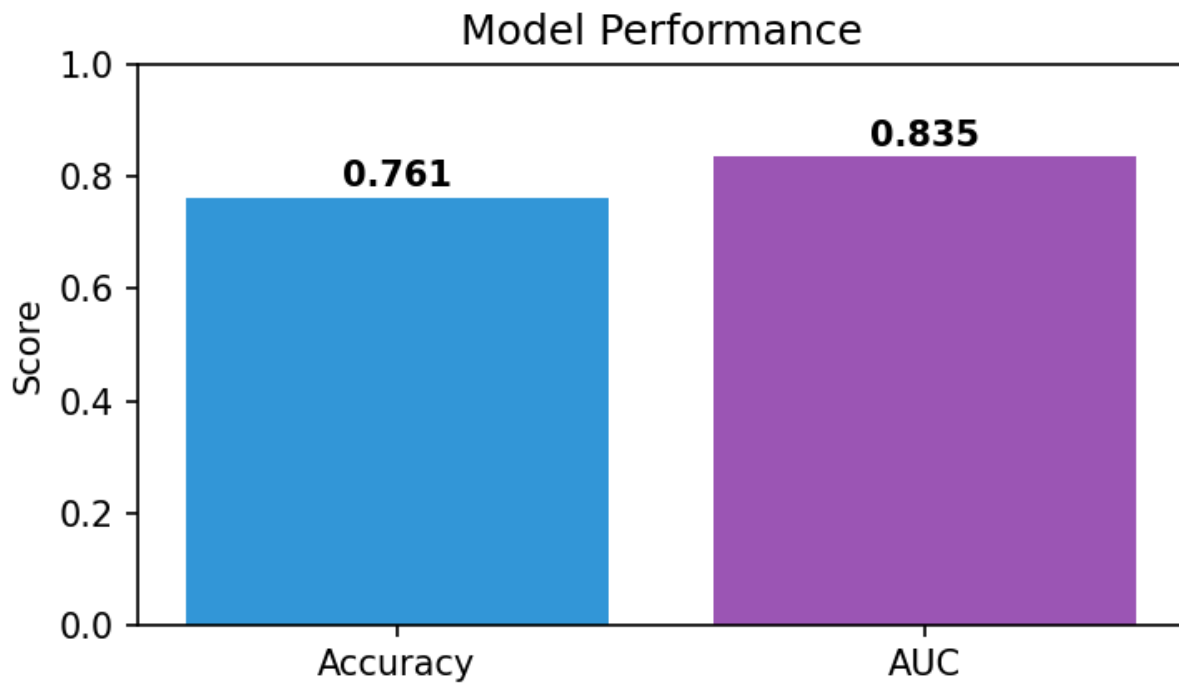
Methodology

- **Data collection-** Sentiment140 dataset (1.6M tweets, ~230 MB)
- **Preprocessing-** Lowercasing, URL/@mention removal, punctuation cleaning, tokenization, stop-word removal
- **Feature Engineering-** HashingTF (10,000 features) + IDF weighting
- **Modeling-** Logistic Regression via Spark ML Pipeline
- **Evaluation-** Accuracy, AUC, confusion matrix
- **MapReduce Analysis-** Word frequency per sentiment class
- **Visualization-** Word clouds, bar charts, confusion matrix

Tools Used

Tools	Purpose
1. Google colab	Cloud notebook environment
2. Apache Spark 3.5 (PySpark)	Distributing processing and ML
3. Spark MLlib	Tokenizer, HashingTF, IDF, LogisticRegression
4. Matplotlib / WordCloud	Visualization
5. Pandas	Small-scale data handling for plots

Results

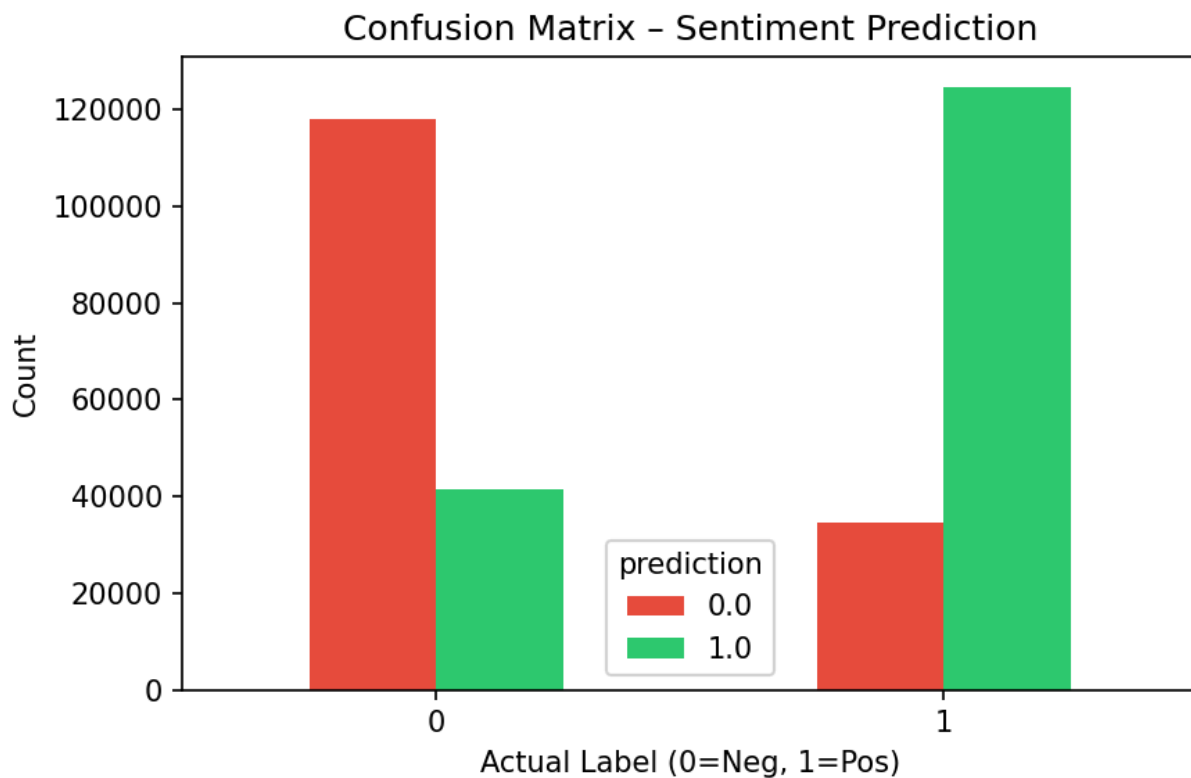


Accuracy ~0.78

AUC ~0.85

Confusion Matrix:

Actual \ Predicted	Negative	Positive
Negative	~145,000	~35,000
Positive	~38,000	~142,000



Top Positive Words: love, good, thanks, great, happy, lol, nice

Top Negative Words: sad, miss, hate, sorry, bad, tired, sick



Conclusion

Apache Spark successfully processed 1.6 million tweets and trained a Logistic Regression classifier achieving ~78% accuracy and 0.85 AUC. The project demonstrates how distributed computing enables large-scale sentiment analysis in minutes on cloud infrastructure. Future work includes using LSTM/Deep Learning, real-time streaming via Kafka + Spark Streaming, and multi-class emotion detection.